Supplementary data

Table S1. Extended table for tDCS exposure for age and population

Study	*n* (total)	*n* (active)	*n* (sham)	*n* (min)	Amperage (mA)	Age (years)	Population	RCT
Mattai (2011) [1]	12 175	5(7) 125	3(2) 50	20 2500	2, sham	10-17 (16.37)	COS	Yes
Faria (2012) [2]	2 6	2 4	2 2	15 60	.5, 1, sham	7-11 (9)	CSWS/LKS	No
Auvichayapat (2013) [3]	36 36	20(7) 27	6(3) 9	20 540	1	6-15 (11.46)	Epilepsy	Yes
Andrade (2013) [4]	14 140	10(4) 140	0	30 4200	2	7-12 (7.57)	ELD/PDD-NOS/AS/GD	No
Gillick (2014) [5]	1 1	1 1	0	10 10	0.7	10	Stroke/Hemi-paretic CP	Yes
Gillick (2015) [6]	11 11	3(2) 5	1(5) 6	10 50	0.7	7-18 (14)	Congenital Hemiparesis	Yes
Moliadze (2015) [7]	19 57	8(11) 38	8(11) 19	10 380	1	11-16 (13.9)	Neurotypical	Yes
Ciechanski (2016) [8]	24 72	11(7) 54	3(3) 18	20 1080	1,2, sham	13-15 (14)	Neurotypical	Yes
Kirton (2017) [9]	23 230	8(4) 120	7(4) 110	20 2400	1, sham	6-18 (11.8)	Unilateral perinatal stroke	Yes
Gómez (2017) [10]	15 300	10(5) 300	0	20 6000	1	5-10 (7.7)	ASD	Yes

Meiron (2017) [11]	1 10	1 10	0	20 200	0.1-1	2.5	Epileptic Encephalopathy	No
Gillick (2018) [12]	20 200	5(5) 100	4(6) 100	20 2000	0.7, sham	7-21 (12.75)	UCP via Hemispheric stroke/PVL	Yes
Rich (2018) [13]	8 80	3(5) 80	0	20 1600	1.5	7-21 (13.4)	Perinatal stroke/UCP	No
Costanzo (2018) [14]	11 198	1(10) 198	0	20 3960	1	10-15 (13.9)	Anorexia Nervosa	No
Cole (2018) [15]	24 120	9(7) 80	2(6) 40	20 1600	1, sham	12-18 (15.5)	Neurotypical	Yes
Costanzo (2019) [16]	26 468	5(8) 234	6(7) 234	20 4680	1, sham	10-17 (13.75)	Dyslexia	Yes
Nemanich (2019) [17]	Same sample as Gillick (2018) [12]				0.7, sham	7-21 (12.75)	UCP via Hemispheric stroke/PVL	Yes
Ciechanski (2019) [18]	1 10	0(1) 10	0	20 200	1.5	17	Stroke Hemiparesis	No
Rahimi (2019) [19]	17 51	9(8) 34	9(8) 17	20 680	1, sham	9-12 (10.35)	Dyslexia	No
Overall	265 2165	111(91) 1560	51(55) 605	32140 535 (h)	0.5, 0.7, 1, 1.5, 2, sham	2.5-21 (12)		

This table summarizes the extent of tDCS exposure for age and population. *n* (total) is the total number of individual participants on top and the total quantity of tDCS sessions below. *n* (active) and *n* (sham) represents the number of males and females (in

parentheses) who received active versus sham tDCS and the number below indicates the quantity of tDCS sessions. *n* (min) indicates the duration of each single tDCS session on top, with the number below representing the number of total minutes of *active* tDCS exposure in each study. Amperage indicates the level of current used in the given sessions. Age is identified first as a range on top and below as an average. Population includes children and adolescents aged 6–18 years (COS = Childhood Onset Schizophrenia; CSWS = Continuous Spikes and Waves during Sleep–Rare Epilepsy; LKS = Landau-Kleffner Syndrome; ELD = Expressive Language Disorder; PDD-NOS = Pervasive Developmental Disorder Not Otherwise Specified; AS = Asperger Syndrome; GP = Global Dyspraxia; CP = Cerebral Palsy; UCP = Unilateral Cerebral Palsy; PVL = Periventricular leukomalacia). RCT indicates if the experiment was a randomized controlled trial.

Table S1 References

1. Mattai, A.; Miller, R.; Weisinger, B.; Greenstein, D.; Bakalar, J.; Tossell, J.; David, C.; Wassermann, E.M.; Rapoport, J.; Gogtay, N. Tolerability of transcranial direct current stimulation in childhood-onset schizophrenia. *Brain Stimul.* **2011**, *4*, 275–280, doi:10.1016/j.brs.2011.01.001.

2. Faria, P.; Fregni, F.; Sebastião, F.; Dias, A.I.; Leal, A. Feasibility of focal transcranial DC polarization with simultaneous EEG recording: Preliminary assessment in healthy subjects and human epilepsy. *Epilepsy Behav.* **2012**, *25*, 417–425, doi:10.1016/j.yebeh.2012.06.027.

3. Auvichayapat, N.; Rotenberg, A.; Gersner, R.; Ngodklang, S.; Tiamkao, S.; Tassaneeyakul, W.; Auvichayapat, P. Transcranial

direct current stimulation for treatment of refractory childhood focal epilepsy. *Brain Stimul.* **2013**, *6*, 696–700, doi:10.1016/j.brs.2013.01.009.

4. Andrade, A.C.; Magnavita, G.M.; Allegro, J.V.B.N.; Neto, C.E.B.P.; Lucena, R. de C.S.; Fregni, F. Feasibility of Transcranial Direct Current Stimulation Use in Children Aged 5 to 12 Years. *J. Child Neurol.* **2013**, *29*, 1369-1365, doi:10.1177/0883073813503710.

5. Gillick, B.T.; Feyma, T.; Menk, J.; Krach, L.E. Safety of transcranial direct current stimulation in pediatric hemiparesis: determination of the method for locating the optimal stimulation site. *Brain Stimul.* **2014**, *7*, e12, doi:10.1016/j.brs.2014.01.042.

6. Gillick, B.T.; Feyma, T.; Menk, J.; Usset, M.; Vaith, A.; Wood, T.J.; Worthington, R.; Krach, L.E. Safety and feasibility of transcranial direct current stimulation in pediatric hemiparesis: randomized controlled preliminary study. *Phys. Ther.* **2015**, *95*, 337–349, doi:10.2522/ptj.20130565.

7. Moliadze, V.; Andreas, S.; Lyzhko, E.; Schmanke, T.; Gurashvili, T.; Freitag, C.M.; Siniatchkin, M. Ten minutes of 1mA transcranial direct current stimulation was well tolerated by children and adolescents: Self-reports and resting state EEG analysis. *Brain Res. Bull.* **2015**, *119*, 25–33, doi:10.1016/j.brainresbull.2015.09.011.

8. Ciechanski, P.; Kirton, A. Transcranial Direct-Current Stimulation Can Enhance Motor Learning in Children. *Cereb. Cortex* **2016**, *43*, bhw114, doi:10.1093/cercor/bhw114.

9. Kirton, A.; Ciechanski, P.; Zewdie, E.; Andersen, J.; Nettel-Aguirre, A.; Carlson, H.; Carsolio, L.; Herrero, M.; Quigley, J.; Mineyko, A.; et al. Transcranial direct current stimulation for children with perinatal stroke and hemiparesis. *Neurology* **2017**, *88*, 259–267, doi:10.1212/WNL.0000000000003518.

10. Gómez, L.; Vidal, B.; Maragoto, C.; Morales, L.M.; Berrillo, S.; Cuesta, H.V.; Baez, M.; Denis, M.; Marín, T.; Cabrera, Y.; et al. Non-invasive brain stimulation for children with autism spectrum disorders: A short-term outcome study. *Behav. Sci. (Basel).* **2017**, *7*, 63, doi:10.3390/bs7030063.

11. Meiron, O.; Gale, R.; Namestnic, J.; Bennet-Back, O.; David, J.; Gebodh, N.; Adair, D.; Esmaeilpour, Z.; Bikson, M. High-Definition transcranial direct current stimulation in early onset epileptic encephalopathy: a case study. *Brain Inj.* **2018**, *32*, 135–143, doi:10.1080/02699052.2017.1390254.

12. Gillick, B.; Rich, T.; Nemanich, S.; Chen, C.Y.; Menk, J.; Mueller, B.; Chen, M.; Ward, M.; Meekins, G.; Feyma, T.; et al.

Transcranial direct current stimulation and constraint-induced therapy in cerebral palsy: A randomized, blinded, sham-controlled clinical trial. *Eur. J. Paediatr. Neurol.* **2018**, *22*, 358–368, doi:10.1016/j.ejpn.2018.02.001.

13. Rich, T.L.; Nemanich, S.; Chen, M.; Friel, K.; Feyma, T.; Krach, L.; Nawshin, T.; Meekins, G.; Gillick, B.T. Transcranial direct current stimulation (tDCS) paired with occupation-centered bimanual training in children with unilateral cerebral palsy: A preliminary study. *Neural Plast.* **2018**, *2018*, 1-14, doi:10.1155/2018/9610812.

14. Costanzo, F.; Menghini, D.; Maritato, A.; Castiglioni, M.C.; Mereu, A.; Varuzza, C.; Zanna, V.; Vicari, S. New Treatment Perspectives in Adolescents With Anorexia Nervosa: The Efficacy of Non-invasive Brain-Directed Treatment. *Front. Behav. Neurosci.* **2018**, *12*, doi:10.3389/fnbeh.2018.00133.

15. Cole, L.; Giuffre, A.; Ciechanski, P.; Carlson, H.L.; Zewdie, E.; Kuo, H.-C.; Kirton, A. Effects of High-Definition and Conventional Transcranial Direct-Current Stimulation on Motor Learning in Children. *Front. Neurosci.* **2018**, *12*, doi:10.3389/fnins.2018.00787.

16. Costanzo, F.; Rossi, S.; Varuzza, C.; Varvara, P.; Vicari, S.; Menghini, D. Long-lasting improvement following tDCS treatment combined with a training for reading in children and adolescents with dyslexia. *Neuropsychologia* **2019**, *130*, 38–43, doi:10.1016/j.neuropsychologia.2018.03.016.

17. Nemanich, S.T.; Rich, T.L.; Chen, C.-Y.; Menk, J.; Rudser, K.; Chen, M.; Meekins, G.; Gillick, B.T. Influence of Combined Transcranial Direct Current Stimulation and Motor Training on Corticospinal Excitability in Children With Unilateral Cerebral Palsy. *Front. Hum. Neurosci.* **2019**, *13*, doi:10.3389/fnhum.2019.00137.

18. Ciechanski, P.; Carlson, H.L.; Herrero, M.; Lane, C.; MacMaster, F.P.; Kirton, A. A Case of Transcranial Direct-Current Stimulation for Childhood Stroke Hemiparesis: A Brief Report. *Dev. Neurorehabil.* **2019**, *23*, 133-136. doi:10.1080/17518423.2019.1655678.

19. Rahimi, V.; Mohamadkhani, G.; Alaghband-Rad, J.; Kermani, F.R.; Nikfarjad, H.; Marofizade, S. Modulation of temporal resolution and speech long-latency auditory-evoked potentials by transcranial direct current stimulation in children and adolescents with dyslexia. *Exp. Brain Res.* **2019**, *237*, 873–882, doi:10.1007/s00221-019-05471-9.

Form S1.

Concomitant Medication Log

CONCOMITANT MEDICATIONS SOURCE DOCUMENT

REB#: **Participant ID#: _____**

Study Name: Transcranial Direct Current Stimulation: A safety study in children and adolescents

Dose Frequency **Route** **Dose Units**

BID = twice daily	Q6h = every 6 hours	QD = once daily	IM – intramuscular	PO – oral	g = gram	mg = milligram	TBSP = tablespoon
TID = 3 times/day	Q8h = every 8 hours	HS = at bedtime	IN – intranasal	SC – subcutaneous	gtt = drop	mL = milliliter	tsp = teaspoon
QID = 4 times/day	Qam = one dose in morning	PRN = as needed	INH – inhaled	TOP – topical	mcg = microgram	oz = ounce	OTH = other, specify
q2h = every 2 hours	Qpm = one dose in evening	OTH = other	IT – intrathecally	OTIC – by ear	mcL = microliter	SPY = spray	UNK = unknown
q4h = every 4 hours		UNK = unknown	IV - intravenous	OTH – other, specify		supp = suppository	

COMPLETE AT INITIATION OF THERAPY									COMPLETE AT DISCONT THERAPY / END O		
MEDICATION / NON-	INDICATION (If given for an	BASELINE MED?	DOSE (indicate	FREQUENCY	ROUTE	START DATE	REPORTER INITIALS	REPORTER DATE	STOP DATE	CONTINUING AT END OF	REPO INIT

MEDICATION THERAPY	AE, enter the exact term detailed on the AE form)	1 – Yes 2 – No	units)			*(yyyy-mmm-dd)*		*(yyyy-mmm-dd*	*(yyyy-mmm-dd)*	**STUDY?** 1 – Yes 2 – No	
1.											
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